

ARTIFICIAL INTELLIGENCE FOR EDUCATORS

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Outline

1 AI DRIVES

About Me

Yogesh Haribhau Kulkarni

Bio:

- ▶ 20+ years in CAD/Engineering software development
- ▶ Got Bachelors, Masters and Doctoral degrees in Mechanical Engineering (specialization: Geometric Modeling Algorithms).
- ▶ Currently doing Coaching in fields such as Data Science, Artificial Intelligence Machine-Deep Learning (ML/DL) and Natural Language Processing (NLP).
- ▶ Feel free to follow me at:
 - ▶ Github (github.com/yogeshhk)
 - ▶ LinkedIn (www.linkedin.com/in/yogeshkulkarni/)
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AI Drives for Educators

Why Now? The Strategic Imperative

- ▶ AI literacy is no longer optional, it is foundational
- ▶ Industry expects AI-literate graduates across all branches
- ▶ Peer institutions are already implementing AI programs
- ▶ Student employability directly linked to AI exposure
- ▶ Delayed action creates widening competitive gap
- ▶ NAAC accreditation may increasingly evaluate AI readiness
- ▶ NEP 2020 emphasizes technology integration across disciplines

Vision and Strategic Approach

- ▶ AI for all disciplines, not only CSE/AI branches
- ▶ Teacher-first multiplier model: empower faculty to scale impact
- ▶ Every teacher uses AI productively and responsibly
- ▶ Every department integrates AI where relevant
- ▶ Students graduate with practical AI exposure
- ▶ Institute becomes recognized AI-enabled ecosystem/campus

Three-Tier AI Development Model

- ▶ Progressive pathway: Literacy → Proficiency → Expertise
- ▶ Level 1: AI Literacy Drive (AILD) - All faculty
- ▶ Level 2: AI Proficiency Drive (AIPD) - Builders
- ▶ Level 3: AI Expertise Drive (AIED) - Researchers
- ▶ Each level has distinct targets and outcomes

Three-Tier AI Development Model

EMPOWER TEACHERS. INSPIRE STUDENTS. BUILD AN AI-READY INSTITUTE.

BUILD AN AI-FIRST CAMPUS. TOGETHER.



Teacher First → Student Multiplier

Empowering teachers with AI skills creates a lasting ripple effect across every classroom and future generation.



1. AI LITERACY DRIVE (AILD)

AI for Every Teacher

For all faculty across departments. Build confidence in using AI tools to enhance teaching, research & productivity.

- ✓ Prompt Engineering Basics
- ✓ AI for Teaching, Research & Administration
- ✓ Responsible & Ethical AI Usage
- ✓ 1-Hour Online Certification



Outcome: Every faculty becomes a confident and responsible AI user.



2. AI PROFICIENCY DRIVE (AIPD)

Build and Apply AI

For CS/IT/AI faculty and interested teachers. Build practical AI applications and solve real institutional & domain problems.

- ✓ RAG Systems & Private Data Chatbots
- ✓ AI Automation & Workflow Tools
- ✓ Hands-on Projects & Hackathons
- ✓ Faculty-Student Innovation Teams



Outcome: Practical AI solutions that create impact across the institute.



3. AI EXPERTISE DRIVE (AIED)

Research and Innovate

For researchers, PG/PhD students & advanced faculty. Drive high-quality research and institutional visibility in AI.

- ✓ Advanced AI Research & Models
- ✓ Publications in Top Conferences & Journals
- ✓ Industry Collaborations & Funded Projects
- ✓ Patents & Open-Source Contributions



Outcome: Research excellence, innovation leadership & global recognition.



Better Student Readiness



More Productive Faculty



Stronger Research Culture



Industry Alignment



Long-term Institutional Differentiation



Let's build AI-ready campuses and shape the future of education!

Connect with me or
workshops & implemen^t

PAIC
Pune AI Community

for more information,
support.



Level 1: AI Literacy Drive (AILD)

- ▶ Target: All faculty across all departments
- ▶ Delivery: Online self-paced certification (1-2 hours) or Youtube + quiz
- ▶ Key capabilities: Prompt engineering, AI-assisted teaching workflows
- ▶ Applications: Lecture prep, assessments, research summaries, reports
- ▶ Critical principle: Never use confidential data in public LLMs
- ▶ Success metrics: 80% faculty certified, one workflow demo per faculty
- ▶ Timeline: Complete baseline certification in 3 months
- ▶ Foundation for institution-wide AI adoption

Level 2: AI Proficiency Drive (AIPD)

- ▶ Target: Technical faculty and student innovation teams
- ▶ Objective: Build institutional AI applications
- ▶ Key technologies: RAG systems, chatbots, workflow automation
- ▶ Sample projects: Admission assistant, library search, placement tools
- ▶ Infrastructure: Shared GPU server, AI sandbox, cloud credits
- ▶ Expected outcome: 5-10 deployed AI applications
- ▶ Move from usage to creation and problem-solving
- ▶ Participation in national AI hackathons and competitions

Level 3: AI Expertise Drive (AIED)

- ▶ Target: Research faculty, PhD scholars, PG students
- ▶ Objective: Build institutional research reputation in AI
- ▶ Focus: Domain-specific AI (healthcare, manufacturing, agriculture)
- ▶ Activities: Research groups, industry collaborations, funded projects
- ▶ Expected outcomes: Scopus/WoS publications, patents, grants
- ▶ Contribution to cutting-edge AI innovation
- ▶ Institutional recognition in national AI research community

Governance and Implementation Structure

- ▶ Internal AI Steering Committee with full ownership of outcomes
- ▶ AI Champions Network: 1-2 champions per department
- ▶ Early adopters serve as mentors and internal trainers
- ▶ Pilot approach: Start with 2-3 departments for refinement
- ▶ Investment: time of the faculty and fees if any paid certification is selected, else none

Implementation Timeline

- ▶ Phase 1 (Months 0-3): Foundation
 - ▶ Form steering committee, identify champions, launch AILD
 - ▶ Target: 80% faculty baseline certification
- ▶ Phase 2 (Months 3-6): Building
 - ▶ Launch AIPD workshops, establish AI lab
 - ▶ Start faculty-student projects, conduct internal hackathon
- ▶ Phase 3 (Months 6-12): Excellence
 - ▶ Launch AIED research clusters, industry partnerships
 - ▶ Deploy institutional AI tools, publish research papers

Expected Outcomes and Impact

- ▶ Month 3: 80% faculty AI-certified across all departments
- ▶ Month 6: 5+ institutional AI applications deployed
- ▶ Month 9: Student AI clubs active, regular hackathons
- ▶ Month 12: 2-3 research publications in AI domain
- ▶ Improved student placement rates and starting salaries
- ▶ Enhanced NAAC scores and institutional rankings
- ▶ Recognition as AI-forward institution in state/region

Approval Request and Next Steps

- ▶ Approve three-tier AI transformation framework
- ▶ Authorize formation of AI Steering Committee
- ▶ Approve faculty release time for AI Champions
- ▶ Grant autonomy for pilot implementation in 2-3 departments
- ▶ Next steps: Committee formation within 2 weeks
- ▶ Launch AILD certification drive within 4 weeks
- ▶ Time to act is now - competitors are not waiting